THE EULER CHARACTERISTIC IS ADDITIVE ON SHORT EXACT SEQUENCES OF COHERENT MODULES ON A PROPER SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof cites no external \ref inside the counted span, so the assembly needs no external-reference stubs.

Allowed standard prerequisites (editorial).

- Finiteness of cohomology of coherent sheaves on a proper scheme. Stacks Project, Cohomology of Schemes (cohomology of coherent sheaves on proper schemes is finite dimensional). Ensures the Euler characteristic is a finite alternating sum, so rank-nullity applies to the long exact sequence.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k. Let $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ be a short exact sequence of coherent modules on X. Then*

$$\chi(X, \mathcal{F}_2) = \chi(X, \mathcal{F}_1) + \chi(X, \mathcal{F}_3)$$

Proof. Consider the long exact sequence of cohomology

$$0 \to H^0(X, \mathcal{F}_1) \to H^0(X, \mathcal{F}_2) \to H^0(X, \mathcal{F}_3) \to H^1(X, \mathcal{F}_1) \to \ldots$$

associated to the short exact sequence of the lemma. The rank-nullity theorem in linear algebra shows that

$$0 = \dim H^0(X, \mathcal{F}_1) - \dim H^0(X, \mathcal{F}_2) + \dim H^0(X, \mathcal{F}_3) - \dim H^1(X, \mathcal{F}_1) + \ldots$$

This immediately implies the lemma. □

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